Richard Harnisch

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Education

MSc Artificial Intelligence 2026–2028
University of Amsterdam

BSc Artificial Intelligence, cum laude, GPA 8.4/10 2022–2026
University of Groningen

- Thesis: Interpolation Threshold and Double Descent in RL. Tracked Fisher Information Matrix trace across over-parameterized agents; found local curvature can decouple from the generalization gap under policy-induced non-stationarity.
- Formal math/CS foundations: Calculus for Artificial Intelligence; Linear Algebra and Multivariable Calculus; Statistics; Analysis; Probability Theory; Probability and Measure; Linear Algebra 2; Partial Differential Equations; Imperative Programming; Algorithms and Data Structures; Object-Oriented Programming.
- Advanced AI/ML: Applied Machine Learning, Reinforcement Learning, Uncertainty in ML, Machine Translation (9.5 each); Neural Networks, Natural Language Processing (9 each); Advanced Logic (8.5).

International Baccalaureate; German Abitur 1.0 2020–2022
Dresden International School 42/45 points

Higher Level Mathematics, Physics, Economics; maximum marks in Mathematics, Physics, German, Spanish. Extended Essay on public-key cryptography including Euler's totient function, Chinese remainder theorem & vulnerabilities.

Research and ML Experience

Research Assistant: Data Center Optimization 2026–present
Multiscale Networked Systems Group Amsterdam

- Working at the University of Amsterdam under Dr. Shashikant Ilager on Data Center design, targeting publication.
- Optimizing electricity use and carbon intensity while keeping performance within threshold values for both inference and training workloads. Exploring gradient descent on data center component specs in simulations and agent-driven design.

Data Science Intern: Electricity Price Forecasting 2025–2026
Nederlandse Gasunie Groningen/Remote

- Developed a day-ahead electricity price forecasting system using SARIMA, LSTMs, and Transformers in 5-person team.
- Designed and deployed distribution-shift monitoring over ENTSO-E price data and Dutch grid-production data; system alerts operators and triggers retraining under covariate drift; worked directly with Chief Data Scientist.
- Conducted feature engineering and data management for real-world forecasting in a grid undergoing renewable- and geopolitics-driven distribution shifts including alerting operator upon distribution shift to retrain model on new data.

Bachelor's Thesis: Double Descent in Reinforcement Learning 2025–2026
University of Groningen Supervisor: Dr. Matthia Sabatelli

- Investigated the interpolation threshold and Double Descent phenomenon in over-parameterized Reinforcement Learning agents under non-stationary data distributions created by randomized environments and adaptive policy.
- Tracked the Fisher Information Matrix (FIM) trace as a generalization metric across constrained policy optimizers (TRPO) at varying model capacity and training durations. Gained experience with high-performance computing systems.
- Found empirical evidence that FIM decorrelates from the generalization gap under policy-induced non-stationarity: constrained optimizers exhibit expected FIM spike near the interpolation threshold, but the curvature signal does not predict generalization gap evolution, challenging the applicability of classical information-geometric bounds in RL.
- Raises open questions about the role of implicit regularization and optimizer geometry in non-stationary generalization.

Machine Learning Engineer and Software Developer 2024–2026
descript GmbH Dresden/Remote

- Independently led industrial ML research collaboration with TU Dresden on textile defect detection; defined project methodology to maximize learning from limited data and built Django interface for uncertainty-prioritzed expert labelling.
- Engineered production features from client requirements, including a scheduling system for hospitality room management.

Teaching Assistant: Introduction to Machine Learning 2025–2026
University of Groningen

- Led lab sessions and graded assignments for 100 second-year AI students, communicated ML concepts to broad audience.
- Topics: ML paradigms, model selection, regularization, evaluation metrics, data ingestion, processing and augmentation.

Selected Projects

Plant Disease Classifier. End-to-end image classification with uncertainty calibration, model compression for edge deployment, ResNets/EfficientNets/SVMs/ensembles, and FastAPI serving; course grade 9.5/10.
Mars Rover Competition. Authored technical report for European Space Agency rover competition and presented with MakerCie team; placed 1st overall among 33 teams and received maximum points for presentation.
Study Association Board. As board of 1,000+ member AI/CS study association at University of Groningen organized AI symposium with academic & industry speakers; ~150 attendees. Organized international excursions to Lisbon & Thessaloniki.

Languages: English (TOEFL iBT 6/6, C2) and German native; Dutch and Spanish conversational.